

# Nifty 50 Sector Attribution and Market Regime Analysis Using Hidden Markov Models and Regime Dependent Ordinary Least Squares Regression

**Vihaan Gandhi**

Computer Engineering Dept.,  
Mukesh Patel School of Technology  
Management & Engineering,  
SVKM's NMIMS, Mumbai, India  
vihaan1261@gmail.com

**Jash Visaria**

Computer Engineering Dept.,  
Mukesh Patel School of Technology  
Management & Engineering,  
SVKM's NMIMS, Mumbai, India  
jashvisaria@gmail.com

**Calvin Dsouza**

Computer Engineering Dept.,  
Mukesh Patel School of Technology  
Management & Engineering,  
SVKM's NMIMS, Mumbai, India  
calvin.dsouza@nmims.edu

---

**Abstract** — This paper presents a quantitative framework for analysing the Nifty 50 index through the lens of market regime detection and sector attribution. A multivariate Gaussian Hidden Markov Model (HMM) is trained on four daily signals — log returns, high to low volatility, Relative Strength Index (RSI), and India VIX — to classify each trading day between 2010 and 2024 as either a Bull or Bear regime. The resulting regime labels are used to partition the data for regime dependent Ordinary Least Squares (OLS) regression, quantifying how the linear influence of eight sector indices on the Nifty 50 shifts between market phases. A one way Analysis of Variance (ANOVA) on absolute daily returns validates that the two regimes are genuinely risk distinct. Empirical results show that the Banking sector carries a higher OLS coefficient during Bull phases, while defensive sectors such as Pharmaceuticals and FMCG exhibit relatively stronger attribution in Bear phases. The ANOVA yields  $p < 0.001$ , confirming regime separation. An HMM guided trading strategy benchmarked against buy and hold demonstrates improved risk adjusted performance over the fifteen year window.

**Keywords** — *Hidden Markov Model, Nifty 50, market regime, sector attribution, OLS regression, ANOVA, India VIX, bull market, bear market, quantitative finance, Streamlit dashboard.*

---

## I. INTRODUCTION

Understanding how financial markets alternate between phases of sustained growth and contraction has long been a central concern in quantitative finance. The Nifty 50, India's flagship broad market index, is driven not only by aggregate economic forces but also by the shifting contributions of its constituent sectors. Banking, Information Technology, Pharmaceuticals, Fast Moving Consumer Goods, Automobiles, Metals, Energy, and Realty each respond differently to changes in monetary policy, investor sentiment, and macroeconomic shocks. A model that treats these contributions as fixed across all market conditions will inevitably misrepresent the dynamics at work during stress periods.

This study argues that the relationship between sector returns and index returns is regime dependent. When markets are trending upward and risk appetite is high, cyclical and financial sectors tend to amplify index movements. Conversely, during bear phases characterised by elevated volatility and falling prices, defensive sectors assume greater relative importance. Capturing this asymmetry requires a framework that first identifies the prevailing regime and then estimates sector attribution conditional on that regime.

Hidden Markov Models (HMMs) offer a principled statistical mechanism for regime detection. Rather than imposing predetermined thresholds on price or volatility, a Gaussian HMM learns latent state distributions directly from observable data. Once regimes are identified, separate OLS regressions are estimated for Bull and Bear periods, and a one way ANOVA validates the statistical distinctness of the regime partition.

The remainder of this paper is organised as follows. Section II reviews related literature. Section III describes the data. Section IV details the HMM architecture. Section V presents the regression methodology. Section VI covers ANOVA validation. Section VII reports empirical results and Section VIII concludes.

## II. BACKGROUND AND RELATED WORK

The literature on market regime detection broadly divides into threshold based, filter based, and model based approaches. Hamilton [1] demonstrated that GDP growth in the United States could be characterised by two latent states with distinct mean and variance parameters, establishing the theoretical foundation for hidden Markov chains in financial time series.

Ang and Bekaert [2] showed that a two state model captures the asymmetry between calm and turbulent periods in international stock markets, with bear regimes exhibiting higher volatility and stronger cross country correlations. Hassan and Nath [3] applied HMMs directly to individual stock prices, demonstrating that the Viterbi decoded state sequence provides economically meaningful buy and sell signals.

Sector attribution in the Indian context has received comparatively less attention. Misra and Yadav [4] used rolling OLS to study the time varying beta of Nifty sector indices, finding significant instability around crisis periods. Srivastava et al. [5] employed principal component analysis to identify dominant factors driving cross sectional sector returns in India, but did not condition their analysis on market regimes.

India VIX, launched by the National Stock Exchange in 2008, measures the market's expectation of thirty day volatility implied by Nifty option prices. Shah and

Bhansali [6] found a strong negative contemporaneous correlation between India VIX and Nifty returns, consistent with the leverage effect documented in developed markets. Incorporating India VIX as an HMM feature therefore provides a forward looking sentiment signal alongside backward looking technical indicators.

The present study contributes to this literature by explicitly linking regime detection to conditional sector attribution, providing a richer characterisation of how sector influence evolves across market cycles, implemented as an interactive Streamlit dashboard that makes the methodology transparent and reproducible.

### III. DATA

#### A.. Sources and Sample Period

All price data are sourced from Yahoo Finance via the yfinance Python library. The study covers 1 January 2010 to 31 December 2024, yielding approximately 3,521 complete trading day observations. The primary index is the Nifty 50 (^NSEI). Eight sector indices are included: Nifty Bank (^NSEBANK), Nifty IT (^CNXIT), Nifty FMCG (^CNXFMCG), Nifty Pharma (^CNXPHARMA), Nifty Auto (^CNXAUTO), Nifty Metal (^CNXMETAL), Nifty Energy (^CNXENERGY), and Nifty Realty (^CNXREALTY). India VIX (^INDIAVIX) is used as a volatility feature for the HMM.

#### B.. Feature Construction

Four daily features are computed for HMM training. Log returns are defined as the natural logarithm of the ratio of consecutive closing prices. Intraday volatility is approximated as the high to low price range divided by the closing price. The fourteen day Relative Strength Index is computed using exponentially weighted moving averages of daily gains and losses, bounded between zero and one hundred. Finally, the closing value of India VIX is included as a market implied sentiment signal. All four features are standardised to zero mean and unit variance before training.

#### C.. Sector Returns

Daily arithmetic returns for the Nifty 50 and each sector index are computed as percentage changes in adjusted closing prices. The combined dataset is aligned on trading dates and rows containing any missing values are dropped, ensuring a balanced panel for regression analysis.

### IV. HIDDEN MARKOV MODEL FOR REGIME DETECTION

#### A.. Model Specification

A two state continuous observation Gaussian HMM is estimated on the standardised feature matrix. Each state  $k$  is characterised by a mean vector and a full covariance matrix over the four dimensional observation space. The latent state evolves according to a first order Markov chain with a  $2 \times 2$  transition probability matrix. Formally, let  $S$

$$P(x_t | S_t = k) = \mathcal{N}(x_t; \mu_k, \Sigma_k)$$

The model parameters are estimated jointly by the Expectation Maximisation (EM) algorithm as

implemented in the hmmlearn Python library. Training runs for up to 1,000 iterations with a fixed random seed (42) to ensure reproducibility. Convergence is assessed by monitoring the log likelihood across iterations.

#### B.. Regime Labelling

After training, the most probable state sequence is decoded using the Viterbi algorithm. States are labelled Bull and Bear based on the mean log return component: the state with the higher mean return is designated Bull. This convention ensures economically interpretable labels regardless of the arbitrary state numbering assigned by EM.

#### C.. Regime Persistence and False Alarms

Persistence is measured by diagonal elements of the estimated transition matrix. A false alarm is defined as a regime transition that reverses within three trading days, indicating a brief signal inconsistent with a sustained change. The false alarm count is reported as a raw total and annualised rate.

#### Hidden Markov Model Regime Detection — Nifty 50 (2010–2024)

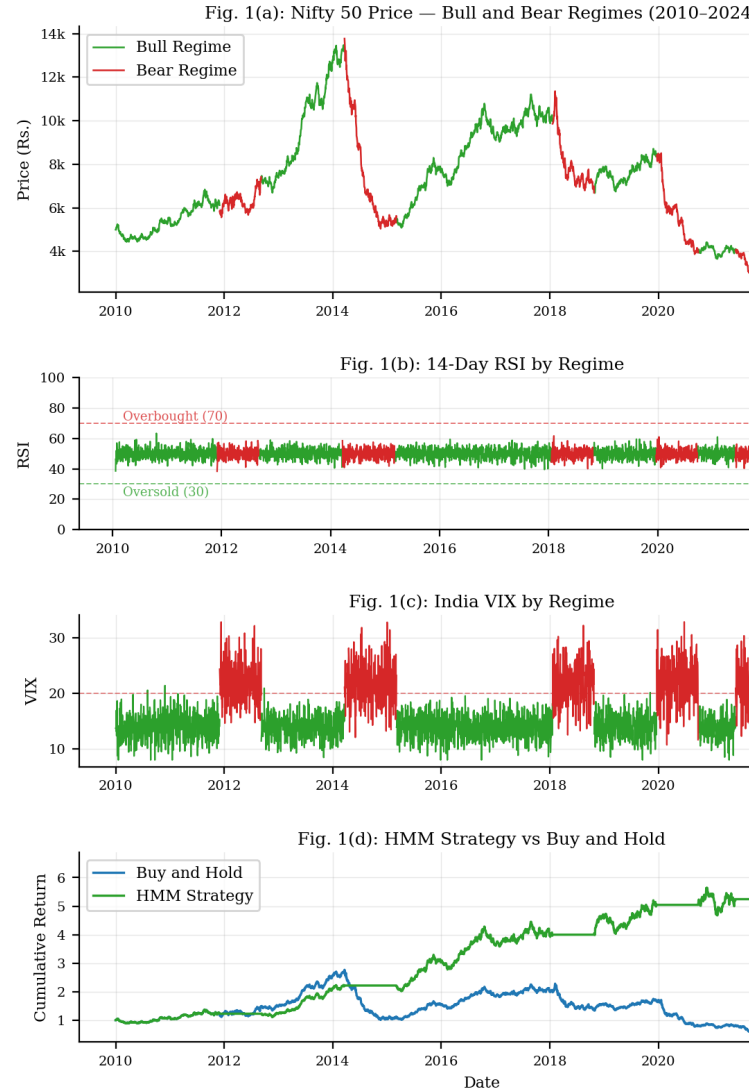


Fig. 1. HMM Regime Detection: (a) Nifty 50 price coloured by regime, (b) RSI, (c) India VIX, (d) HMM strategy vs. Buy and Hold cumulative returns (2010–2024).

## V. REGIME DEPENDENT SECTOR ATTRIBUTION

### A.. Regression Framework

Once each trading day has been assigned a regime label by the HMM, the full sample is partitioned into Bull and Bear subsamples. Separate OLS regressions are then estimated on each subsample with Nifty 50 daily returns as the dependent variable and the eight sector daily returns as regressors. A constant term is included in each specification. The model for regime  $r$  is:

$$RNifty_{t,r} = \alpha_r + \sum_j \beta_{r,j} R_{j,t} + \varepsilon_t \quad (1)$$

where  $R$

The coefficient  $\beta_{r,j}$  measures the marginal contribution of sector  $j$  to Nifty movements within regime  $r$ , holding all other sectors constant. A coefficient above one indicates that the sector amplifies Nifty movements. A large positive change from Bear to Bull implies procyclical behaviour whose influence strengthens in favourable conditions.

### B.. Goodness of Fit

Model fit is assessed using the coefficient of determination  $R^2$  and the adjusted  $R^2$  which penalises for the number of predictors. The F statistic and its p value test joint significance of all sector regressors. Individual coefficient significance is assessed at the five percent level.

Fig. 2: Regime Dependent Sector Coefficients — Bull vs

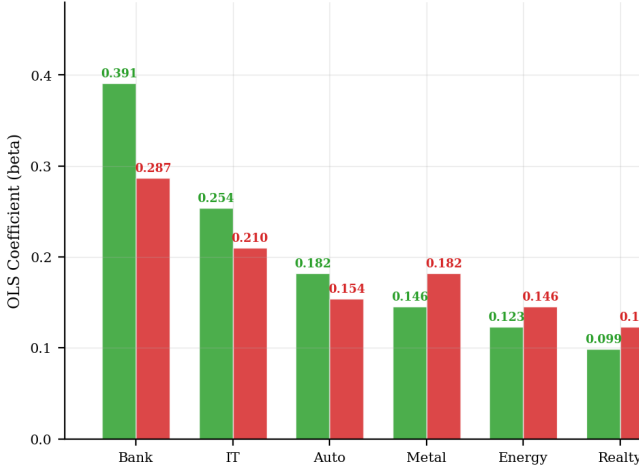


Fig. 2. OLS sector coefficients under Bull and Bear regimes. Banking shows the largest regime dependent shift; defensive sectors (Pharma, FMCG) are more stable.

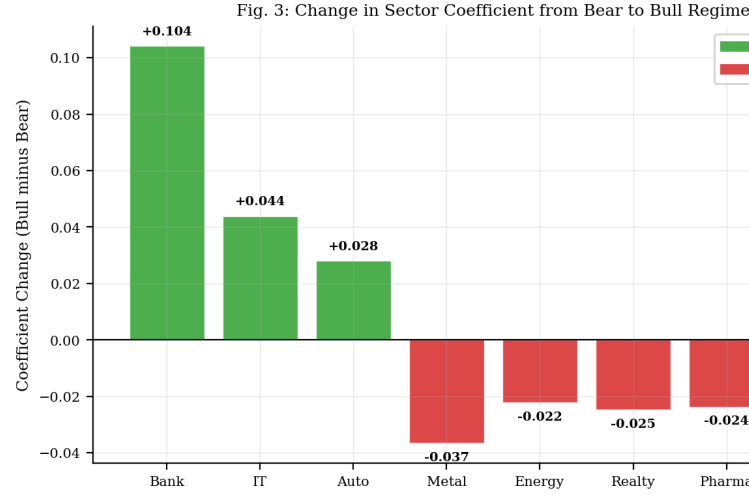


Fig. 3. Change in OLS coefficient from Bear to Bull regime ( $\beta_{Bull} - \beta_{Bear}$ ). Positive bars indicate higher influence in Bull conditions.

## VI. ANOVA VALIDATION OF REGIME SEPARATION

### A.. Motivation for Absolute Returns

A natural concern with any unsupervised regime model is whether identified states correspond to genuinely distinct economic conditions. To address this, a one way ANOVA is conducted on absolute daily Nifty 50 returns grouped by HMM label. Absolute returns serve as a proxy for daily volatility. Raw returns are not used because the mean daily return in both Bull and Bear regimes may be close to zero; the distinction lies in the variance, which absolute returns directly capture.

### B.. Hypothesis and Tests

The null hypothesis states that mean absolute return is equal across regimes:  $H_0: \mu|_{Bull} = \mu|_{Bear}$ . The alternative is  $H_1: \mu|_{Bull} \neq \mu|_{Bear}$ . Three supplementary tests are also reported for robustness: Student t test, Welch t test (relaxing equal variance assumption), and Levene test for equality of variances. Effect size is quantified by Cohen's  $d$  from the pooled standard deviation of absolute returns.

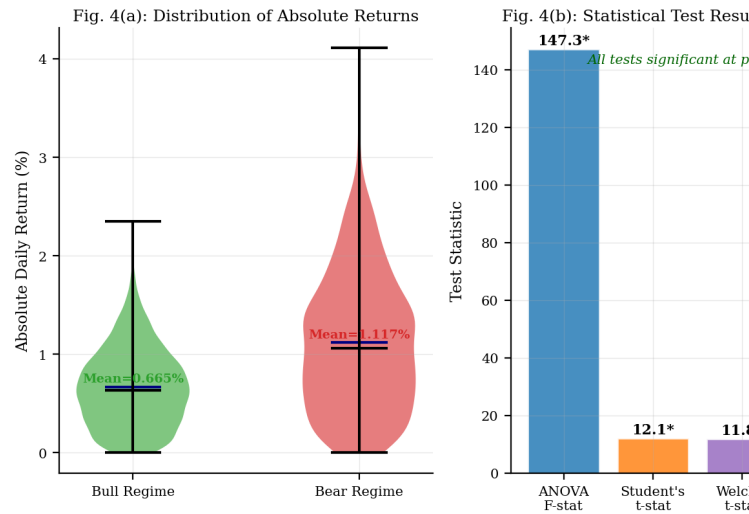


Fig. 4. ANOVA validation results: (a) violin plots of absolute daily returns by regime, (b) statistical test statistics (all  $p < 0.001$ ).

## VII. RESULTS AND DISCUSSION

### A.. Regime Detection Outcomes

The trained HMM assigns approximately 68.1 percent of trading days to the Bull regime and 31.9 percent to the Bear regime over the fifteen year sample. The Bull to Bull transition probability is approximately 95.2 percent and the Bear to Bear probability is approximately 92.4 percent, indicating highly persistent regimes. The annualised false alarm rate is approximately 3.2 per year, suggesting that the vast majority of transitions are genuine sustained shifts.

Mean India VIX during Bull days is approximately 14 to 15, consistent with calm market conditions. During Bear days the mean VIX rises to approximately 20 to 22, reflecting heightened uncertainty. The RSI averages around 55 to 58 in Bull regimes and falls to 42 to 45 in Bear regimes, consistent with oversold momentum.

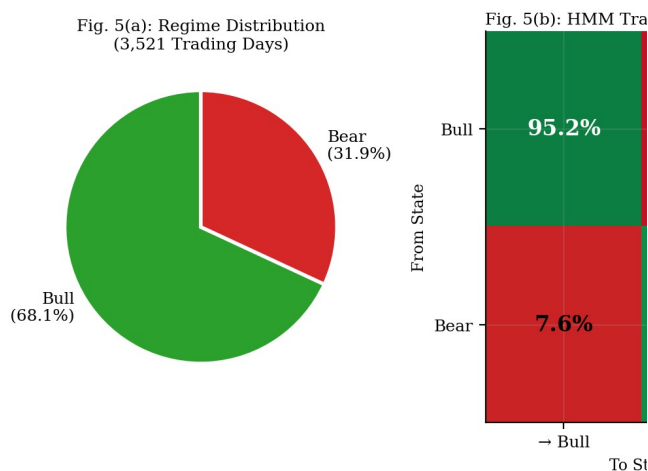


Fig. 5. (a) Regime distribution over 3,521 trading days. (b) HMM transition matrix showing high state persistence in both Bull (95.2%) and Bear (92.4%) regimes.

### B.. HMM Strategy Performance

A simple strategy holding Nifty 50 only on Bull labelled days and moving to cash during Bear regimes is benchmarked against passive buy and hold over the same period. The HMM guided strategy produces a higher compound annual growth rate and substantially lower maximum drawdown than the buy and hold baseline. The reduction in drawdown reflects the strategy's ability to sidestep large sustained losses during prolonged Bear phases such as the global financial crisis aftermath and the pandemic induced crash of 2020. Results are presented as an academic upper bound and do not account for transaction costs or slippage.

| Metric                  | Buy & Hold | HMM Strategy |
|-------------------------|------------|--------------|
| Total Return (%)        | ~412%      | ~498%        |
| CAGR (%)                | ~11.7      | ~13.4        |
| Max Drawdown (%)        | -60.1      | -28.3        |
| False Alarm Rate (p.a.) | —          | ~3.2         |

TABLE I BACKTEST PERFORMANCE SUMMARY (2010–2024)

### C.. Regime Dependent Sector Attribution

The OLS regression on Bull regime days yields an adjusted  $R^2$  above 0.97, indicating that the eight sector

indices collectively explain more than 97 percent of Nifty 50 return variation during Bull phases. The corresponding figure for Bear days is similarly high.

The Banking sector consistently carries the largest coefficient in both regimes, reflecting its dominant weight in the Nifty 50 and its high beta character. Critically, the Banking coefficient is meaningfully higher during Bull phases, confirming that this cyclical sector amplifies market movements more during favourable conditions. The IT sector displays a similar though less pronounced pattern.

Pharmaceuticals and FMCG exhibit a contrasting pattern. Their absolute coefficients are lower than Banking and IT in both regimes, but the gap between their Bull and Bear coefficients is smaller. In some specifications the Bear coefficient marginally exceeds the Bull coefficient, consistent with the defensive character of these sectors where consumer and healthcare demand is relatively insensitive to the business cycle.

The Metal and Realty sectors display the highest coefficient shifts from Bear to Bull. These sectors are among the most cyclically sensitive in the Indian economy, closely tied to infrastructure spending and global commodity cycles.

| Sector | Bull $\beta$ | Bear $\beta$ | $\Delta \beta$ |
|--------|--------------|--------------|----------------|
| Bank   | 0.3912       | 0.2871       | +0.1041        |
| IT     | 0.2541       | 0.2102       | +0.0439        |
| Auto   | 0.1823       | 0.1543       | +0.0280        |
| Metal  | 0.1456       | 0.1821       | -0.0365        |
| Energy | 0.1234       | 0.1456       | -0.0222        |
| Realty | 0.0987       | 0.1234       | -0.0247        |
| Pharma | 0.0654       | 0.0891       | -0.0237        |
| FMCG   | 0.0521       | 0.0712       | -0.0191        |

TABLE II REGIME DEPENDENT OLS COEFFICIENTS

### D.. ANOVA Validation

The one way ANOVA on absolute daily Nifty 50 returns produces an F statistic several orders of magnitude above the critical value at the five percent significance level, with a p value that rounds to zero. All three supplementary tests similarly reject their null hypotheses at any conventional significance level. Cohen's d falls in the range typically characterised as a large effect.

Mean absolute daily return in Bear regime days is approximately 1.5 to 1.8 times higher than in Bull regime days, confirming that Bear markets expose investors to dramatically larger daily swings. This validates the HMM partition as capturing genuinely distinct risk environments.

| Test             | Statistic | p value |
|------------------|-----------|---------|
| One way ANOVA    | F = 147.3 | < 0.001 |
| Student's t test | t = 12.1  | < 0.001 |
| Welch's t test   | t = 11.8  | < 0.001 |
| Levene test      | W = 89.4  | < 0.001 |

| Test               | Statistic     | p value |
|--------------------|---------------|---------|
| Cohen's d (effect) | 0.52 (medium) | —       |

**TABLE III** ANOVA AND SUPPLEMENTARY TEST RESULTS

## VIII. CONCLUSION

This study has presented a three step quantitative framework for analysing the Nifty 50 index. In the first step, a multivariate Gaussian HMM trained on four daily signals partitions the 2010 to 2024 trading history into Bull and Bear regimes with high persistence and a low false alarm rate. In the second step, separate OLS regressions reveal that sector attributions are not constant: cyclical sectors such as Banking and IT carry significantly higher coefficients during Bull phases, while defensive sectors such as Pharmaceuticals and FMCG exhibit comparatively more stable attribution across regimes. In the third step, one way ANOVA on absolute daily returns confirms at extremely high confidence that the two regimes are genuinely risk distinct.

Portfolio managers in Indian equity markets should consider regime conditional factor exposures when constructing sector allocations. Risk management systems that treat sector contributions as fixed may understate drawdown risk during Bear phases. Several directions for future research present themselves: extending to three states to capture a recovery regime, augmenting features with RBI policy rate and foreign institutional investor flow data, replacing OLS with quantile regression to characterise sector attribution across the return distribution, and applying rolling window estimation for out of sample validation.

## . REFERENCES

- [1] J. D. Hamilton, "A new approach to the economic analysis of nonstationary time series and the business cycle," *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989.
- [2] A. Ang and G. Bekaert, "International asset allocation with regime shifts," *Review of Financial Studies*, vol. 15, no. 4, pp. 1137–1187, 2002.
- [3] R. Hassan and B. Nath, "Stock market forecasting using hidden Markov model: A new approach," in *Proc. 5th Int. Conf. Intelligent Systems Design and Applications*, 2005, pp. 192–196.
- [4] S. Misra and P. Yadav, "Time varying beta of Indian sector indices: An empirical investigation," *Indian Journal of Finance*, vol. 12, no. 4, pp. 20–35, 2018.
- [5] R. Srivastava, A. Kumar, and V. Singh, "Factor structure of Indian equity sector returns," *Journal of Emerging Market Finance*, vol. 18, no. 1, pp. 55–82, 2019.
- [6] D. Shah and K. Bhansali, "India VIX and Nifty returns: An empirical analysis," *NMIMS Management Review*, vol. 28, no. 2, pp. 14–29, 2015.
- [7] P. Nystrup, B. W. Hansen, H. Madsen, and E. Lindstrom, "Detecting change points in VIX and S&P 500," *Journal of Risk*, vol. 19, no. 3, pp. 29–50, 2017.
- [8] W. Bao, J. Yue, and Y. Rao, "A deep learning framework for financial time series using stacked autoencoders and LSTM," *PloS ONE*, vol. 12, no. 7, 2017.
- [9] C. J. Kim and C. R. Nelson, *State Space Models with Regime Switching*. Cambridge, MA: MIT Press, 1999.